Total-variation contraction of nonlinear recombination for equal one-site marginals

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Editorial note. The counted item is Theorem3.2 for two initial distributions sharing every one-site marginal. The nearby equilibrium/entropy corollary434-443 and1269-1288 is not a second counted problem.

Editorial note. The pointers to equations3.50 and3.53 only invite comparison with a similar calculation later in the paper. The actual Wild recursion, marked-tree product representation and exact tree-weight ODE needed here are all retained as proof obligations.

Editorial note. Author notation at1207-1212 writes a two-element set in V_j; it denotes inclusion of both coordinates, consistently with the event |V_j|>=2 and the separation definition196-198. Original mathematical text is retained.

Editorial note. The native source1172-1179 contains a left delimiter across an aligned display; any necessary delimiter-size/layout adaptation must retain the same scalar-product difference and be documented in assembly, not treated as a new mathematical step.

Editorial scope. One nonlinear same-marginal contraction theorem, with its full local construction and tree expectation proof. The equilibrium relative-entropy corollary is not separately counted; old particle entropy-production theorem remains a different published item.

Allowed standard prerequisites (editorial).

- Finite-dimensional probability ODEs and uniqueness. Interpret the recombination semigroup; the Wild expansion and tree representation are proved local ingredients, not supplied conclusions.
- Total-variation optimal coupling and dual characterization. TV distance equals minimum mismatch probability and one half of the sup over bounded test functions of norm<=1.. Couple independent leaf configurations and convert their mismatch estimate to TV.
- Nonnegative series and elementary integral/ODE manipulation. Sum the finite-tree renewal identity and solve its resulting scalar linear ODE.

Source reference. Original equation 3.50: sources/nonlinear-recombinations-ahl-2024-219/source0.tex, source lines 1373–1396. The text explicitly says similar computation; actual nonlinear recursion is supplied964-965 and its direct verification is elementary. No estimate from the later equation is assumed.

Source reference. Original equation 3.53: sources/nonlinear-recombinations-ahl-2024-219/source0.tex, source lines 1399–1408. The current tree induction is stated using its explicit recursion975 and product1077. This pointer supplies an analogous worked calculation, not a missing result used in the proof.

0.1 The nonlinear equation

Let $\Omega = \prod_{i=1}^{n} X_i$ be the set of n-vectors $\sigma = (\sigma_1, \ldots, \sigma_n)$ where $\sigma_i \in X_i$, and the X_i are given finite sets. We interpret σ as a particle. Thus, a particle is a string of n characters each taken from a finite alphabet. A basic example is obtained by taking $\Omega = \{0,1\}^n$. Without loss of generality we will assume that each X_i has the form $X_i := \{0, 1, 2, \ldots, q_i\}$, for some $q_i \in \mathbb{N}$. Given a subset $A \subset [n]$, $[n] = \{1, \ldots, n\}$, and $\sigma \in \Omega$, σ_A denotes the $A-$component of σ, that is the string $(\sigma_i, i \in A)$. If $(\sigma, \eta) \in \Omega \times \Omega$ is a pair of particles, the recombination at A consists in exchanging the A-component of σ with the A-component of η. This defines the map

$$(\sigma, \eta) \mapsto (\eta_A \sigma_{A^c}, \sigma_A \eta_{A^c}),$$

where $\eta_A \sigma_{A^c}$ denotes the element of Ω with entries η_i for $i \in A$ and σ_i for $i \in A^c = [n] \setminus A$. Let $\mathcal{P}(\Omega)$ denote the set of probability measures on Ω. If the original pair (σ, η) is obtained by sampling independently from $p \in \mathcal{P}(\Omega)$, then the new particle $\eta_A \sigma_{A^c}$ is distributed according to $p_A \otimes p_{A^c}$, the product of the two marginals of p on the A and A^c components respectively. By choosing the set $A \subset [n]$ according to some distribution ν, one obtains the quadratic collision kernel

$$p \mapsto Q(p) = \sum_{A \subset [n]} \nu(A)\, p_A \otimes p_{A^c}.$$

The nonlinear evolution is defined by the dynamical system $\dot{p}_t = Q(p_t) - p_t$, that is

$$\frac{d}{dt}\, p_t = \sum_{A \subset [n]} \nu(A)\, (p_{t,A} \otimes p_{t,A^c} - p_t) \tag{1}$$

with the initial condition $p_0 = p \in \mathcal{P}(\Omega)$. Here $p_t \in \mathcal{P}(\Omega)$ is the distribution of the particle at time t, $p_{t,A}$ denotes its marginal on A and ν is a given probability measure over the subsets of $[n]$. The study of this model starts with the pioneering work of Geiringer [21]; see also [29, 39, 38, 40] for more recent accounts. It is well known that the Cauchy problem associated to (1) has a unique solution for every initial distribution $p \in \mathcal{P}(\Omega)$; see e.g. [40]. Moreover, it is not difficult to see that the evolution preserves the single site marginals, that is $p_{t,i} = p_i$ for all $t \geqslant 0$ and for all $i \in [n]$. We say that the recombination measure ν is *separating* if for any $i, j \in [n]$ there is a positive probability that the random set A with distribution ν separates i and j. Equivalently, if $r(\nu) < 1$ where

$$r(\nu) := \max_{i < j \in [n]} \nu\left(\{i,j\} \subset A \text{ or } \{i,j\} \subset A^c\right) \tag{2}$$

0.2 Contractive estimates for the nonlinear semigroup

It is convenient to use the notation $S_t(p) = p_t$ for the solution of (6) with initial datum $p_0 = p$, so that $S_{t+s} = S_t S_s = S_s S_t$, for all $s, t \geqslant 0$. This defines the nonlinear semigroup $\{S_t,\, t \geqslant 0\}$. We show a contraction property for S_t when the two initial distributions p, q have the same marginals, that is $p_i = q_i$ for all $i \in [n]$.

Theorem 0.1. *For any probability measure $p, q \in \mathcal{P}(\Omega)$ such that $p_i = q_i$ for all $i \in [n]$, all $t \geqslant 0$,*

$$\|S_t(p) - S_t(q)\|_{\mathrm{TV}} \leqslant \tfrac{1}{2}\, n(n-1)\, \|p - q\|_{\mathrm{TV}}\, e^{-D(\nu)t} \tag{3}$$

where $D(\nu) = 1 - r(\nu)$.

0.3 Wild sums and McKean trees

Given $f, g : \Omega \to \mathbb{R}$, we adopt the notation

$$(f \circ g)^A := \frac{1}{2}\left(f_A \otimes g_{A^c} + g_A \otimes f_{A^c}\right), \tag{4}$$

$$f \circ g := \sum_{A \subset [n]} \nu(A)(f \circ g)^A, \tag{5}$$

where $f_A(\sigma_A) := \sum_{\sigma_{A^c}} f(\sigma_A \sigma_{A^c})$. Then the nonlinear equation (1) is given by

$$\frac{d}{dt}\, p_t = p_t \circ p_t - p_t. \tag{6}$$

The convolution product defined by $f \circ g$ is commutative and distributive, but not associative. Following Wild's original construction [44] we write the solution of (6) with initial datum $p_0 = p$, as

$$p_t = e^{-t} \sum_{k=1}^{\infty} \left(1 - e^{-t}\right)^{k-1} p^{(k)}, \tag{7}$$

where the $\{p^{(k)}\}_{k \,\geqslant\, 1}$ are probability measures on Ω defined inductively by

$$p^{(1)} = p, \qquad p^{(k)} = \frac{1}{k-1} \sum_{j=1}^{k-1} p^{(j)} \circ p^{(k-j)}\,, \quad k \geqslant 2. \tag{8}$$

The validity of (7) can be easily checked by direct inspection, see e.g. the argument after (3.50) below for a similar computation.

Moreover, following McKean [31], we may express $p^{(k)}$ as a weighted sum over rooted binary trees γ with k leaves. We now recall the details of this construction, and refer the reader to [7] for further background. Let $\Gamma(k)$ denote the set of rooted binary trees with k leaves and call $\alpha_k(\gamma)$, $\gamma \in \Gamma(k)$, the probability over $\Gamma(k)$ obtained by the following procedure. $\Gamma(1) = \{\gamma_1\}$ is just the empty tree with only the root with $\alpha_1(\gamma_1) = 1$, and $\Gamma(2) = \{\gamma_2\}$ where γ_2 is the unique tree obtained by adding two children to the root, with $\alpha_2(\gamma_2) = 1$. Then, recursively, for $k \geqslant 2$, for any $\gamma_{k-1} \in \Gamma(k-1)$, consider all possible trees

$\gamma_{k-1}^i \in \Gamma(k)$, $i = 1, \ldots, k-1$, obtained by adding two children to the ith leaf of γ_{k-1}, and for any $\gamma \in \Gamma(k)$, set

$$\alpha_k(\gamma) = \sum_{\gamma_{k-1} \in \Gamma(k-1)} \alpha_{k-1}(\gamma_{k-1}) \sum_{i=1}^{k-1} \frac{\mathbf{1}\left(\gamma = \gamma_{k-1}^i\right)}{k-1}\,.$$

This defines a probability α_k on $\Gamma(k)$, for any $k \in \mathbb{N}$, and one checks, recursively, that for all $k \geqslant 2$,

$$\alpha_k(\gamma) = \frac{1}{k-1}\alpha_j(\gamma_l)\alpha_{k-j}(\gamma_r), \tag{9}$$

where γ_l and γ_r denote respectively the subtree of γ rooted at the left child of the root and the subtree of γ rooted at the right child of the root, while j denotes the number of leaves in γ_l. Then, by induction over k it follows that for all $k \in \mathbb{N}$,

$$p^{(k)} = \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) C_\gamma(p), \tag{10}$$

where $C_\gamma(p) \in \mathcal{P}(\Omega)$ is described as follows. Each internal node of γ represents a collision and the tree γ describes the collision history. Then $C_\gamma(p)$ represents the distribution obtained at the root after all collisions from γ have been performed, starting with the distribution p at each leaf of γ.

For example, if $k = 3$, there are only two trees $\gamma, \gamma' \in \Gamma(3)$, with $\alpha_3(\gamma) = \alpha_3(\gamma') = 1/2$ and

$$C_\gamma(p) = (p \circ p) \circ p = p \circ (p \circ p) = C_{\gamma'}(p). \tag{11}$$

We refer to Figure 1 for two examples of $C_\gamma(p)$ for $\gamma \in \Gamma(4)$.

Since each collision is of the form $q \circ q'$ for some $q, q' \in \mathcal{P}(\Omega)$ and since by (4) one has

$$p \circ q = \sum_{A \subset [n]} \bar{\nu}(A)\, p_A \otimes q_{A^c}, \tag{12}$$

where $\bar{\nu}(A) = \frac{1}{2}(\nu(A) + \nu(A^c))$, one can write the following expansion for the resulting measure at the root:

$$C_\gamma(p) = \sum_{\vec{A} \in \mathcal{V}_n^{k-1}} \nu(\vec{A}) C_\gamma^{\vec{A}}(p), \tag{13}$$

where $\mathcal{V}_n$ is the set of subsets of $[n]$, so that $\vec{A} \in \mathcal{V}_n^{k-1}$ represents a pattern $\vec{A} = (A_1, \ldots, A_{k-1})$,

$$\nu(\vec{A}) = \prod_{i=1}^{k-1} \bar{\nu}(A_i) \tag{14}$$

and $C_\gamma^{\vec{A}}(p) \in \mathcal{P}(\Omega)$ is the distribution computed as follows. Each internal node v_i, $i = 1, \ldots, k-1$, in γ (in some fixed order) is associated with the set A_i, and we attach the mark A_i to the edge connecting v_i to its left child and the mark A_i^c to the edge connecting v_i to its right child in γ. In this way we obtain

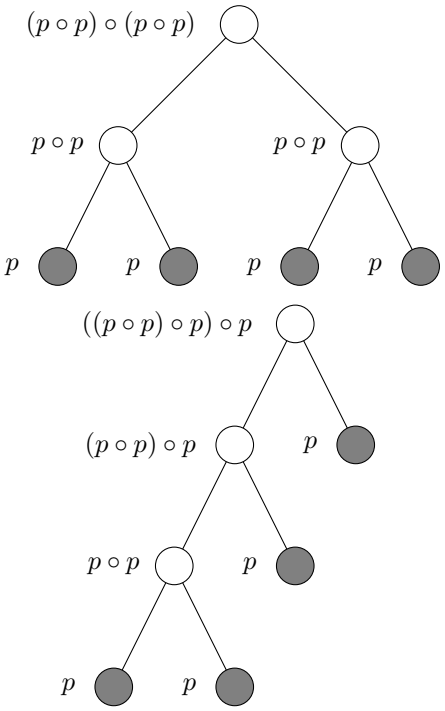


Figure 1: Two possible trees $\gamma, \gamma' \in \Gamma(4)$, and the corresponding distributions $C_\gamma(p) = (p \circ p) \circ (p \circ p)$ and $C_{\gamma'}(p) = ((p \circ p) \circ p) \circ p$.

a tree γ with marks on all its edges. Let d_i denote the depth of the ith leaf of γ (e.g. counting from the leftmost leaf), and consider $p_{V_i(\vec{A})}$, the marginal of the measure p on the subset

$$V_i(\vec{A}) := \cap_{j=1}^{d_i} A_j^i, \tag{15}$$

where $A_1^i, \ldots, A_{d_i}^i$ are the marks encountered along the edges of the unique path from the root to the ith leaf. Then $C_\gamma^{\vec{A}}(p)$ is given by

$$C_\gamma^{\vec{A}}(p) = p_{V_1(\vec{A})} \otimes \cdots \otimes p_{V_k(\vec{A})}. \tag{16}$$

Note that some of the $V_i(\vec{A})$ may be empty. However, they form a partition of $[n]$, namely $V_i(\vec{A}) \cap V_j(\vec{A}) = \emptyset$ for $i \neq j$ and $\cup_{i=1}^k V_i(\vec{A}) = [n]$, which can be seen as the result of a fragmentation process

$$[n] \to (A_1, A_1^c) \to (A_1 \cap A_2, A_1 \cap A_2^c, A_1^c) \to \cdots \to \left(V_1(\vec{A}), \cdots, V_k(\vec{A})\right).$$

As an example, in the case (11) we have $\vec{A} = (A_1, A_2)$, and

$$(p \circ p) \circ p = \sum_{A_1,\, A_2 \subset [n]} \bar{\nu}(A_1)\bar{\nu}(A_2) p_{A_1 \cap A_2} \otimes p_{A_1 \cap A_2^c} \otimes p_{A_1^c}. \tag{17}$$

The proof of (10)-(13) can be easily done by induction, by splitting the tree $\gamma \in \Gamma(k)$ into the left and right subtrees γ_l, γ_r and using the relation (9), see the proof of (3.53) below for a closely related explicit computation.

In conclusion, from (7), (10) and (13) we obtain the following representation of the distribution p_t as a convex combination of distributions $C_\gamma^{\vec{A}}(p)$:

$$p_t = \sum_{k=1}^{\infty} \beta_t(k) \sum_{\gamma \,\in\, \Gamma(k)} \alpha_k(\gamma) \sum_{\vec{A} \,\in\, \mathcal{V}_n^{k-1}} \nu(\vec{A})\, C_\gamma^{\vec{A}}(p), \tag{18}$$

where $\beta_t(k) := e^{-t}(1 - e^{-t})^{k-1}$ is a probability on $\mathbb{N}$ for each $t \geqslant 0$, $\alpha_k(\gamma)$ is a probability on $\Gamma(k)$ for all k, and $\nu(\vec{A})$ is a probability on $\mathcal{V}_n^{k-1}$ for all k.

Proof. From (18) we write

$$S_t(p) - S_t(q) = \sum_{k=1}^{\infty} \beta_t(k) s^{(k)}, \tag{19}$$

where

$$s^{(k)} = \sum_{\gamma \,\in\, \Gamma(k)} \alpha_k(\gamma) \sum_{\vec{A} \,\in\, \mathcal{V}_n^{k-1}} \nu(\vec{A}) \left[C_\gamma^{\vec{A}}(p) - C_\gamma^{\vec{A}}(q)\right]. \tag{20}$$

We introduce some further notation in order to handle more explicitly the difference of probability measures $C_\gamma^{\vec{A}}(p) - C_\gamma^{\vec{A}}(q)$. For any given $\gamma \in \Gamma(k)$, and $\vec{A} \in \mathcal{V}_n^{k-1}$, $\phi : \Omega \mapsto \mathbb{R}$, we may write

$$\begin{aligned} &\langle C_\gamma^{\vec{A}}(p) - C_\gamma^{\vec{A}}(q), \phi \rangle \\ &\quad = \sum_{(x,y) \,\in\, \Omega^k \,\times\, \Omega^k} \mu(x,y) \sum_{z \,\in\, \Omega} \phi(z) \left(\mathbf{1}\left(z = u(y, \vec{A})\right) - \mathbf{1}\left(z = u(x, \vec{A})\right)\right) \end{aligned} \tag{21}$$

where $x = (x^i, i = 1, \ldots, k) \in \Omega^k$ stands for the configurations sampled with $q^{\otimes k}$ over the leaves of γ, $y = (y^i, i = 1, \ldots, k) \in \Omega^k$ stands for the configurations sampled with $p^{\otimes k}$ over the leaves of γ, and $\mu(x, y) = \prod_{i=1}^k \mu_1(x^i, y^i)$ denotes a coupling of these two product measures, such that for every $i = 1, \ldots, k$, $\mu_1(x^i, y^i)$ is the optimal coupling of (q, p):

$$\|p - q\|_{\mathrm{TV}} = \sum_{(x,y) \in \Omega^k \times \Omega^k} \mu(x, y) \mathbf{1}\left(x^i \neq y^i\right) = \sum_{x^i, y^i \in \Omega} \mu_1\left(x^i, y^i\right) \mathbf{1}\left(x^i \neq y^i\right). \tag{22}$$

The notation $u(x, \vec{A})$ in (21) is defined as follows. Note that $x^i \in \Omega$ is a vector $x^i = (x^i_1, \ldots, x^i_n)$, for every i, with $x^i_\ell \in \{0, \ldots, q_\ell\}$, and write $x^i_A = (x^i_\ell)_{\ell \in A}$ for the content of x^i on $A \subset [n]$. With this notation, given $x \in \Omega^k$ and $\vec{A} \in \mathcal{V}_n^{k-1}$, $u(x, \vec{A}) \in \Omega$ is defined as the unique configuration such that

$$u(x, \vec{A})_{V_i(\vec{A})} = x^i_{V_i(\vec{A})}$$

for every $i = 1, \ldots, k$. In words, for each $i \in [k]$, the content of the configuration $u(x, \vec{A})$ on the set $V_i(\vec{A}) \subset V$ is taken from the configuration x^i at the ith leaf. The validity of (21) is thus a consequence of (16).

Let $\partial\gamma$ denote the set of leaves of γ. Notice that

$$\mathbf{1}(z = u(x, \vec{A})) = \prod_{i \in \partial\gamma} \mathbf{1}(z_{V_i(\vec{A})} = x^i_{V_i(\vec{A})})$$

Let $F = F(\vec{A}, \gamma) \subset \partial\gamma$ denote the set of leaves $i \in \partial\gamma$ such that $|V_i(\vec{A})| > 1$, and write

$$\mathbf{1}(z = u(x, \vec{A})) = w_F(x, z, \vec{A}) w_{F^c}(x, z, \vec{A}),$$

where for any $S \subset \partial\gamma$ we write

$$w_S(x, z, \vec{A}) = \prod_{i \in S} \mathbf{1}(z_{V_i(\vec{A})} = x^i_{V_i(\vec{A})}). \tag{23}$$

We write $X = (X^i)_{i=1,\ldots,k}$ and $Y = (Y^i)_{i=1,\ldots,k}$ for the random variables with distribution $q^{\otimes k}$ and $p^{\otimes k}$ respectively. Using the fact that μ is a product over the leaves, and the fact that p, q have the same marginals, for fixed $z \in \Omega$ we have

$$\begin{aligned} \mu\left[w_{F^c}\left(X, z, \vec{A}\right)\right] &= \sum_{(x,y)} \mu(x, y) \prod_{i \in F^c} \mathbf{1}\left(z_{V_i(\vec{A})} = x^i_{V_i(\vec{A})}\right) \\ &= \sum_{(x,y)} \mu(x, y) \prod_{i \in F^c} \mathbf{1}\left(z_{V_i(\vec{A})} = y^i_{V_i(\vec{A})}\right) = \mu\left[w_{F^c}(Y, z, \vec{A})\right], \end{aligned} \tag{24}$$

and

$$\begin{aligned} \mu\left[w_{\partial\gamma}\left(Y, z, \vec{A}\right)\right] &= \sum_{(x,y)} \mu(x, y) \mathbf{1}\left(z = u(y, \vec{A})\right) \\ &= \mu\left[w_{F^c}\left(X, z, \vec{A}\right)\right] \mu\left[w_F\left(Y, z, \vec{A}\right)\right]. \end{aligned} \tag{25}$$

With this notation we rewrite (21) as

$$\begin{aligned}\left\langle C_\gamma^{\vec{A}}(p) - C_\gamma^{\vec{A}}(q), \phi \right\rangle \\ &= \sum_{z\,\in\,\Omega} \phi(z) \left[\mu[w_{\partial\gamma}\left(Y,z,\vec{A}\right)] - \mu[w_{\partial\gamma}\left(X,z,\vec{A}\right)\right], \\ &= \sum_{z\,\in\,\Omega} \phi(z) \mu\left[w_{F^c}\left(X,z,\vec{A}\right)\right]\, \mu\left[w_F\left(Y,z,\vec{A}\right) - w_F\left(X,z,\vec{A}\right)\right].\end{aligned} \tag{26}$$

Note that $w_F(Y,z,\vec{A})] \neq w_F(X,z,\vec{A})$ implies that there exists $i \in F$ such that $X^i \neq Y^i$. Therefore, from (22) we see that

$$\begin{aligned}&\sum_{z\,\in\,\Omega} \mu\left[w_{F^c}\left(X,z,\vec{A}\right)\right]\, \left|\mu\left[w_F\left(Y,z,\vec{A}\right) - w_F\left(X,z,\vec{A}\right)\right]\right| \\ &\leqslant \sum_{i\,\in\,F}\sum_{z\,\in\,\Omega} \mu\left[w_{F^c}\left(X,z,\vec{A}\right)\right]\, \mu\left[\left|w_F\left(Y,z,\vec{A}\right) - w_F\left(X,z,\vec{A}\right)\right| \mathbf{1}\left(X^i \neq Y^i\right)\right] \\ &\leqslant 2\sum_{i\,\in\,F} \mu\left[X^i \neq Y^i\right] = 2|F|\,\|p-q\|_{\mathrm{TV}}.\end{aligned} \tag{27}$$

Therefore, we have shown that (20) satisfies

$$\left|\left\langle s^{(k)}, \phi\right\rangle\right| \leqslant 2\,\|\phi\|_\infty\,\|p-q\|_{\mathrm{TV}} \sum_{\gamma\,\in\,\Gamma(k)} \alpha_k(\gamma) \sum_{\vec{A}\,\in\,\mathcal{V}_n^{k-1}} \nu(\vec{A})\,|F|. \tag{28}$$

To estimate (28), let us call $\mathcal{A}_j = \mathcal{A}_j(\gamma)$ the set of $\vec{A} \in \mathcal{V}_n^{k-1}$ such that $|V_j(\vec{A})| \geqslant 2$, and let $\mathcal{A} = \cup_{j=1}^k \mathcal{A}_j$. If $\vec{A} \notin \mathcal{A}$ then all sets $V_j(\vec{A})$ are either empty or a single site and therefore $|F| = 0$. Moreover, $|F| = \sum_{j=1}^k \mathbf{1}(\mathcal{A}_j)$. Therefore we write

$$\sum_{\vec{A}\,\in\,\mathcal{V}_n^{k-1}} \nu(\vec{A})\,|F| = \sum_{j=1}^k \nu(\mathcal{A}_j). \tag{29}$$

To estimate the probability $\nu(\mathcal{A}_j)$, observe that the event $\mathcal{A}_j$ implies that there exists $i_1 < i_2 \in [n]$ such that $\{i_1,i_2\} \in V_j(\vec{A})$. Moreover, recalling (15),

$$\nu\left(\{i_1,i_2\} \in V_j(\vec{A})\right) = \nu\left(\{i_1,i_2\} \in \cap_{l=1}^{d_j} A_l^j\right) = \bar{\nu}\left(\{i_1,i_2\} \in A\right)^{d_j} \leqslant \left(\frac{r(\nu)}{2}\right)^{d_j},$$

where we use the fact that for any $i_1 < i_2$, the probability under $\bar{\nu}$ that $\{i_1,i_2\} \in A$ is bounded by $r(\nu)/2$. Indeed, conditionally on not being separated the probability that both i_1, i_2 belong to A under $\bar{\nu}$ is $1/2$ by symmetry. A union bound then shows that

$$\nu(\mathcal{A}_j) \leqslant \tfrac{1}{2}\, n(n-1) \left(\frac{r(\nu)}{2}\right)^{d_j}.$$

In conclusion, if we define

$$\omega(\gamma) := \sum_{j=1}^k \left(\frac{r(\nu)}{2}\right)^{d_j}, \tag{30}$$

we obtain, for all $k \in \mathbb{N}$,

$$\left|\left\langle s^{(k)}, \phi \right\rangle\right| \leqslant n(n-1)\, \|\phi\|_\infty\, \|p-q\|_{\mathrm{TV}} \sum_{\gamma\, \in\, \Gamma(k)} \alpha_k(\gamma)\, \omega(\gamma). \tag{31}$$

From [7, Lemma 1.4], one has, for all $\varepsilon > 0$,

$$\sum_{\gamma\, \in\, \Gamma(k)} \alpha_k(\gamma)\omega(\gamma) \leqslant B_\varepsilon e^{-a_\varepsilon \log k}. \tag{32}$$

with $a_\varepsilon = (1 - r(\nu))/(1+\varepsilon)$ and $B_\varepsilon = B_\varepsilon(r(\nu), \varepsilon)$. One could use this estimate to obtain an almost optimal decay rate a_ε. However, here we obtain the optimal exponential decay rate by computing the expectation of $\omega(\gamma)$.

Lemma 0.2. *For any $t \geqslant 0$,*

$$\sum_{k=1}^{\infty} \beta_t(k) \sum_{\gamma\, \in\, \Gamma(k)} \alpha_k(\gamma)\, \omega(\gamma) = e^{-(1-r(\nu))t}. \tag{33}$$

Assuming the validity of (33), and using $\|\mu - \mu'\|_{\mathrm{TV}} = \frac{1}{2} \max_{f:\, \|f\|_\infty \leqslant 1} |\mu(f) - \mu'(f)|$, from (19) and (31) we obtain

$$\|S_t(p) - S_t(q)\|_{\mathrm{TV}} \leqslant \tfrac{1}{2}\, n(n-1)\, \|p-q\|_{\mathrm{TV}}\, e^{-(1-r(\nu))t}, \tag{34}$$

which concludes the proof of Theorem 0.1.

It remains to prove (33). For any $\gamma \in \cup_{k=1}^{\infty} \Gamma(k)$, define $\mathbb{P}_t(\gamma) = \beta_t(k)\alpha_k(\gamma)$, where k is such that $\gamma \in \Gamma(k)$. Notice that for any $\gamma \in \cup_{k=1}^{\infty}\Gamma(k)$,

$$\mathbb{P}_t(\gamma) = \mathbf{1}_{\gamma = \emptyset}\, e^{-t} + \mathbf{1}_{\gamma \neq \emptyset} \int_0^t e^{-s} \mathbb{P}_{t-s}(\gamma_l) \mathbb{P}_{t-s}(\gamma_r) ds, \tag{35}$$

where $\emptyset$ denotes the empty tree with one leaf (given by the root), while for $\gamma \neq \emptyset$ we write γ_l, γ_r for the left and right subtrees. Indeed, for $\gamma = \emptyset$ (35) is immediate, while for $\gamma \neq \emptyset$, supposing $\gamma \in \Gamma(k)$, and recalling (9) one has

$$\begin{aligned}
\mathbb{P}_t(\gamma) &= e^{-t}\left(1 - e^{-t}\right)^{k-1} \alpha_k(\gamma) = e^{-t} \int_0^t e^{-(t-s)} \left(1 - e^{-(t-s)}\right)^{k-2} (k-1)\alpha_k(\gamma) ds \\
&= \int_0^t e^{-s} e^{-2(t-s)} \left(1 - e^{-(t-s)}\right)^{k-2} \alpha_j(\gamma_l)\alpha_{k-j}(\gamma_r) ds = \int_0^t e^{-s} \mathbb{P}_{t-s}(\gamma_l)\mathbb{P}_{t-s}(\gamma_r) ds.
\end{aligned}$$

To prove Lemma 0.2, note that the left hand side of (33) is given by $\sum_\gamma \mathbb{P}_t(\gamma)\omega(\gamma)$. Using (35) and $\omega(\emptyset) = 1$, we see that

$$\begin{aligned}
\zeta(t) :&= \sum_\gamma \mathbb{P}_t(\gamma)\omega(\gamma) = e^{-t} + \frac{r(\nu)}{2} \sum_{\gamma \neq \emptyset} \int_0^t e^{-s} \mathbb{P}_{t-s}(\gamma_l)\mathbb{P}_{t-s}(\gamma_r)(\omega(\gamma_l) + \omega(\gamma_r)) ds \\
&= e^{-t} + r(\nu) \sum_\gamma \int_0^t e^{-s} \mathbb{P}_{t-s}(\gamma)\omega(\gamma) ds = e^{-t} + r(\nu) \int_0^t e^{-s} \zeta(t-s) ds.
\end{aligned}$$

Differentiating, and integrating by parts the resulting expression one finds $\dot{\zeta}(t) = -(1 - r(\nu))\zeta(t)$. Since $\zeta(0) = 1$ we obtain $\zeta(t) = e^{-(1-r(\nu))t}$, $t \geqslant 0$. □

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